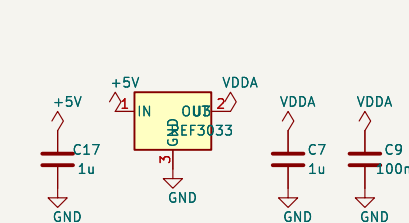
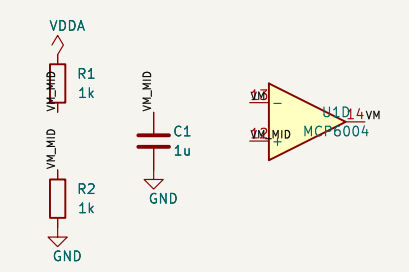


LSU 4.9 analog front-end – topology after rusEFI/FOME wideband module rev C (MIT). See FRONTEND.md

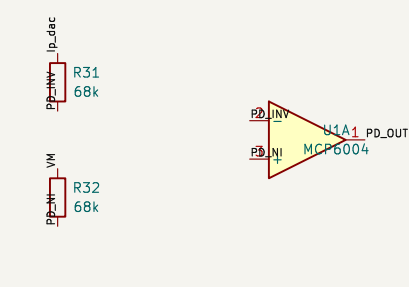
Analog reference VDDA = 3.3 V



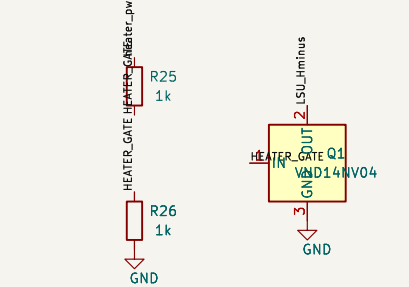
Virtual ground VM = VDDA/2



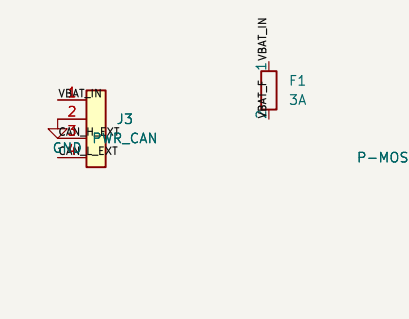
Pump driver: Howland current source, $I = -(R33/R31) \cdot (I_{p_dac} - VM) / R_{pump}$ pump current sense: $I_{p_sense} = VM + 10 \cdot (V_{lp} - V_{Rtrim})$, shunt 61.9 ohm



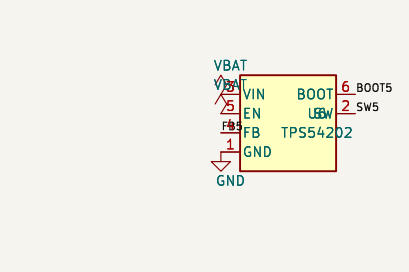
Heater switch



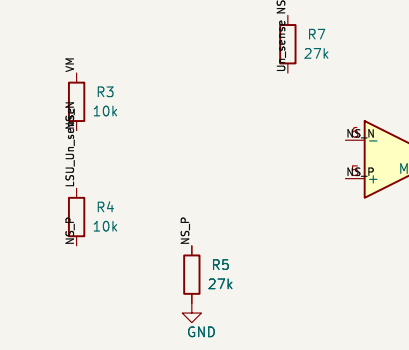
Power input: fuse, reverse-polarity P-MOSFET, TVS



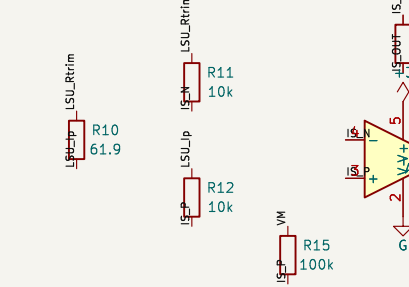
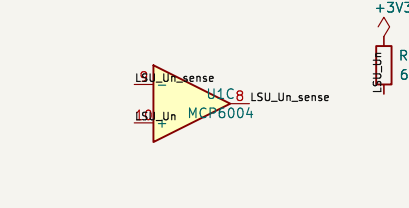
5 V buck (TPS54202: pinout and values from memory – VERIFY against datasheet) 3.3 V LDO



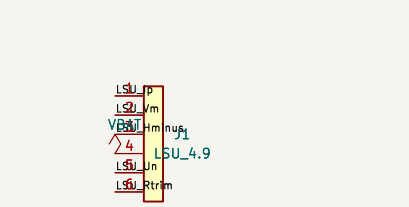
Nernst sense: $Un_sense = 2.7 \cdot (Un - VM)$, target $Un - VM = 0.45 V$



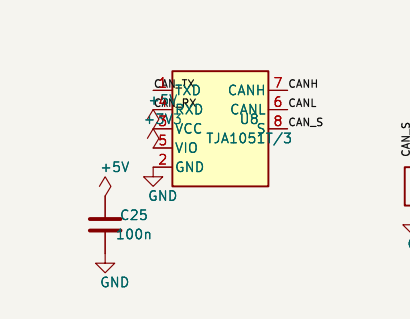
Nernst buffer, bias current (62k) and Ri injection (22k + 100n)



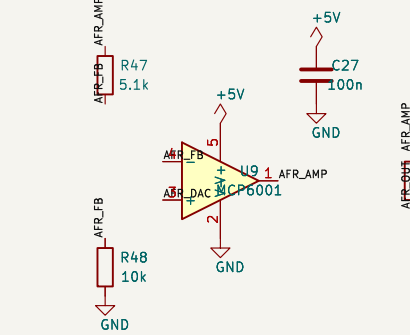
LSU 4.9 connector (pin order to be verified against the Bosch datasheet)



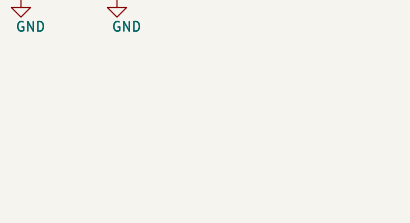
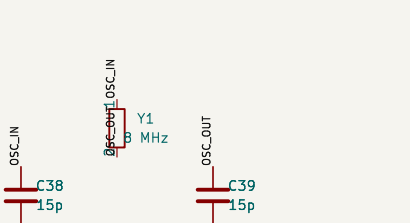
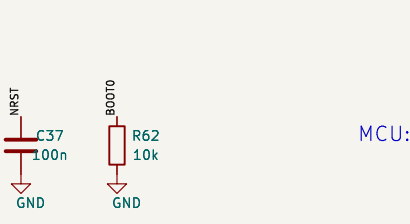
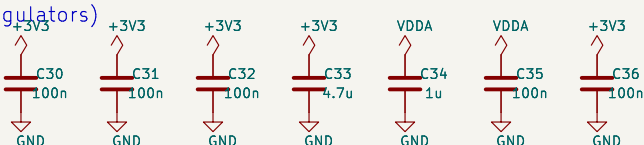
CAN (rusEFI wideband protocol)



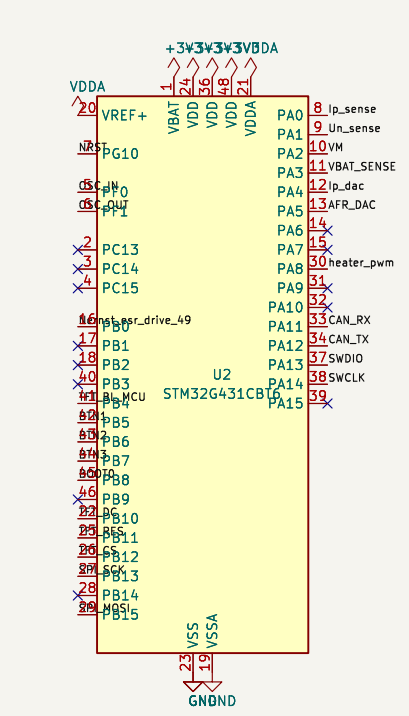
0–5 V AFR output: 10..20 AFR = 0.5 V, DAC 0–3.3 V x1.51



Power flags (ERC: nets fed from connectors / regulators)



MCU: STM32G431CBT6 (pin functions from memory – verify against datasheet) +FFemidAFe i(6C70891XSPI), buttons, SWD



Draft – NOT verified by ERC/simulation/bench
Topology after rusEFI/FOME wideband module rev C (MIT)